

Администрирование сетевых подсистем

Лабораторная работа №8 — Настройка SMTP-сервера на базе Postfix

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Цели и задачи работы

Цель лабораторной работы

Приобрести практические навыки установки и конфигурирования SMTP-сервера на базе Postfix в виртуальной среде, а также проверить доставку локальной и доменной почты.

- Установить и запустить Postfix на server и client
- Выполнить настройку параметров Postfix через postconf
- Проверить доставку локальной почты и пересылку между узлами
- Настроить доставку на доменные адреса с использованием DNS (A/MX/PTR)
- Подготовить provisioning-скрипты для автоматизации развёртывания

Выполнение лабораторной работы

Установка Postfix на сервере

```
[frank@server ~]$ sudo -i
```

```
We trust you have received the usual lecture from the local System
Administrator. It usually boils down to these three things:
```

- #1) Respect the privacy of others.
- #2) Think before you type.
- #3) With great power comes great responsibility.

```
[sudo] password for frank:
```

```
[root@server ~]# dnf -y install postfix
```

```
Extra Packages for Enterprise Linux 9 - x86_64 6.8 kB/s | 37 kB 00:05
Extra Packages for Enterprise Linux 9 - x86_64 1.1 MB/s | 20 MB 00:18
Extra Packages for Enterprise Linux 9 openh264 167 B/s | 993 B 00:05
Rocky Linux 9 - BaseOS 285 B/s | 4.1 kB 00:14
Rocky Linux 9 - BaseOS 151 kB/s | 2.5 MB 00:17
Rocky Linux 9 - AppStream 413 B/s | 4.5 kB 00:11
Rocky Linux 9 - AppStream 497 kB/s | 9.5 MB 00:19
Rocky Linux 9 - Extras 260 B/s | 2.9 kB 00:11
Rocky Linux 9 - Extras 1.1 kB/s | 17 kB 00:15
Dependencies resolved.
```

```
=====
Package           Architecture Version           Repository      Size
=====
Installing:
postfix            x86_64        2:3.5.25-1.el9   appstream      1.5 M
```

```
Transaction Summary
```

```
=====
Install 1 Package
```

```
Total download size: 1.5 M
```

```
Installed size: 4.4 M
```

Настройка параметров myorigin и mydomain

```
complete:
[root@server ~]# dnf -y install s-nail
Last metadata expiration check: 0:05:25 ago on Wed 22 Oct 2025 09:32:43 PM UTC.
Dependencies resolved.
=====
Package                Architecture          Version              Repository           Size
=====
Installing:
s-nail                 x86_64                14.9.22-6.el9       appstream            621 k
Transaction Summary
=====
Install 1 Package

Total download size: 621 k
Installed size: 1.1 M
Downloading Packages:
s-nail-14.9.22-6.el9.x86_64.rpm                                109 kB/s | 621 kB    00:05
-----
Total                                                            53 kB/s | 621 kB    00:11
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      : 1/1
  Running scriptlet: s-nail-14.9.22-6.el9.x86_64 1/1
  Installing     : s-nail-14.9.22-6.el9.x86_64 1/1
  Running scriptlet: s-nail-14.9.22-6.el9.x86_64 1/1
  Verifying      : s-nail-14.9.22-6.el9.x86_64 1/1

Installed:
```

Рис. 2: Изменение параметра myorigin

Отключение IPv6 в Postfix

```
[root@server ~]# firewall-cmd --add-service=smtp
success
[root@server ~]# firewall-cmd --add-service=smtp --permanent
success
[root@server ~]# firewall-cmd --list-services
cockpit dhcpv6-client smtp ssh
[root@server ~]# restorecon -vR /etc
Relabeled /etc/sysconfig/network-scripts/ifcfg-eth1 from unconfined_u:object_r:user_tmp_t:s0 to unconfined_u:object_r:net_conf_t:s0
[root@server ~]# systemctl enable postfix
Created symlink /etc/systemd/system/multi-user.target.wants/postfix.service → /usr/lib/systemd/system/postfix.service.
[root@server ~]# systemctl start postfix
[root@server ~]#
```

Рис. 3: Отключение IPv6

Проверка локальной доставки письма

```
[root@server ~]# postconf myorigin
myorigin = $myhostname
[root@server ~]# postconf mydomain
mydomain = frank.net
[root@server ~]# postconf -e 'myorigin = $mydomain'
postconf: fatal: missing '=' after attribute name: "???myorigin"
[root@server ~]# postconf myorigin
myorigin = $myhostname
[root@server ~]#
```

Рис. 4: Просмотр содержимого почтового ящика

Анализ журнала maillog

```
[root@server ~]# postconf -n
alias_database = hash:/etc/aliases
alias_maps = hash:/etc/aliases
command_directory = /usr/sbin
compatibility_level = 2
daemon_directory = /usr/libexec/postfix
data_directory = /var/lib/postfix
debug_peer_level = 2
debugger_command = PATH=/bin:/usr/bin:/usr/local/bin:/usr/X11R6/bin ddd $daemon_directory/$process_name
                    $process_id & sleep 5
html_directory = no
inet_interfaces = localhost
inet_protocols = all
mail_owner = postfix
mailq_path = /usr/bin/mailq.postfix
manpage_directory = /usr/share/man
meta_directory = /etc/postfix
mydestination = $myhostname, localhost.$mydomain, localhost
mydomain = frank.net
newaliases_path = /usr/bin/newaliases.postfix
queue_directory = /var/spool/postfix
readme_directory = /usr/share/doc/postfix/README_FILES
sample_directory = /usr/share/doc/postfix/samples
sendmail_path = /usr/sbin/sendmail.postfix
setgid_group = postdrop
shlib_directory = /usr/lib64/postfix
smtp_tls_CAfile = /etc/pki/tls/certs/ca-bundle.crt
```

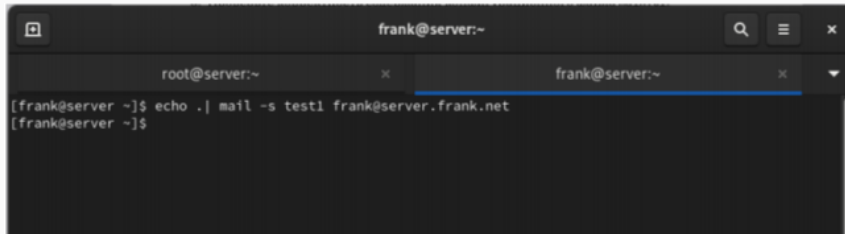
Рис. 5: Анализ журнала maillog

Настройка Postfix на клиенте

```
[root@server ~]# postconf -e 'mydomain = frank.net'
[root@server ~]# postconf inet_protocols
inet_protocols = all
[root@server ~]# postconf -e 'inet_protocols = ipv4'
[root@server ~]# postfix check
[root@server ~]# systemctl reload postfix
[root@server ~]#
```

Рис. 6: Настройка Postfix на клиенте

Разрешение пересылки между узлами

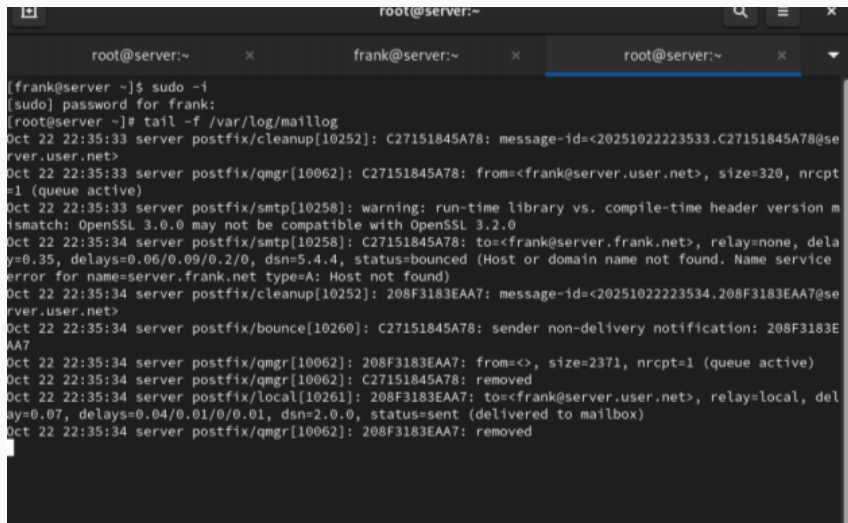


A terminal window titled 'frank@server:~' with search, menu, and close icons. It contains two tabs: 'root@server:~' and 'frank@server:~'. The 'frank@server:~' tab is active and shows the command `echo . | mail -s test1 frank@server.frank.net` being executed. The prompt `[frank@server ~]$` appears on both lines.

```
frank@server:~  
[frank@server ~]$ echo . | mail -s test1 frank@server.frank.net  
[frank@server ~]$
```

Рис. 7: Настройка Postfix на сервере

Подтверждение доставки по логам



The screenshot shows a terminal window with three tabs: 'root@server:~', 'frank@server:~', and 'root@server:~'. The active tab is 'root@server:~'. The terminal content shows a user 'frank' running 'sudo -i' to become root, then running 'tail -f /var/log/maillog'. The log output shows several messages, including a bounced email from 'frank@server.user.net' to 'frank@server.frank.net' due to a host not found error, and a successful local delivery of a message to 'frank@server.user.net'.

```
root@server:~  
[frank@server ~]$ sudo -i  
[sudo] password for frank:  
[root@server ~]# tail -f /var/log/maillog  
Oct 22 22:35:33 server postfix/cleanup[10252]: C27151845A78: message-id=<20251022223533.C27151845A78@se  
rver.user.net>  
Oct 22 22:35:33 server postfix/qmgr[10062]: C27151845A78: from=<frank@server.user.net>, size=320, nrcpt  
=1 (queue active)  
Oct 22 22:35:33 server postfix/smtp[10258]: warning: run-time library vs. compile-time header version m  
ismatch: OpenSSL 3.0.0 may not be compatible with OpenSSL 3.2.0  
Oct 22 22:35:34 server postfix/smtp[10258]: C27151845A78: to=<frank@server.frank.net>, relay=none, dela  
y=0.35, delays=0.06/0.09/0.2/0, dsn=5.4.4, status=bounced (Host or domain name not found. Name service  
error for name=server.frank.net type=A: Host not found)  
Oct 22 22:35:34 server postfix/cleanup[10252]: 208F3183EAA7: message-id=<20251022223534.208F3183EAA7@se  
rver.user.net>  
Oct 22 22:35:34 server postfix/bounce[10260]: C27151845A78: sender non-delivery notification: 208F3183E  
AA7  
Oct 22 22:35:34 server postfix/qmgr[10062]: 208F3183EAA7: from=<>, size=2371, nrcpt=1 (queue active)  
Oct 22 22:35:34 server postfix/qmgr[10062]: C27151845A78: removed  
Oct 22 22:35:34 server postfix/local[10261]: 208F3183EAA7: to=<frank@server.user.net>, relay=local, del  
ay=0.07, delays=0.04/0.01/0/0.01, dsn=2.0.0, status=sent (delivered to mailbox)  
Oct 22 22:35:34 server postfix/qmgr[10062]: 208F3183EAA7: removed
```

Рис. 8: Анализ журнала maillog

Ошибка при отправке на доменный адрес

```
#1) Respect the privacy of others.
#2) Think before you type.
#3) With great power comes great responsibility.

[sudo] password for frank:
[root@client ~]# dnf -y install postfix
Last metadata expiration check: 0:04:58 ago on Wed 22 Oct 2025 11:11:26 PM UTC.
Dependencies resolved.
=====
Package           Architecture Version           Repository        Size
=====
Installing:
postfix            x86_64          2:3.5.25-1.el9    appstream         1.5 M
Transaction Summary
=====
Install 1 Package

Total download size: 1.5 M
Installed size: 4.4 M
Downloading Packages:
postfix-3.5.25-1.el9.x86_64.rpm          6.2 MB/s | 1.5 MB    00:00
=====
```

Рис. 9: Ошибка доставки доменной почты

Проверка очереди сообщений на клиенте

```
#1) Respect the privacy of others.
#2) Think before you type.
#3) With great power comes great responsibility.

[sudo] password for frank:
[root@client ~]# dnf -y install postfix
Last metadata expiration check: 0:04:58 ago on Wed 22 Oct 2025 11:11:26 PM UTC.
Dependencies resolved.
=====
Package                Architecture  Version           Repository        Size
=====
Installing:
postfix                 x86_64        2:3.5.25-1.el9    appstream         1.5 M
Transaction Summary
=====
Install 1 Package

Total download size: 1.5 M
Installed size: 4.4 M
Downloading Packages:
postfix-3.5.25-1.el9.x86_64.rpm        6.2 MB/s | 1.5 MB    00:00
```

Рис. 10: Очередь сообщений Postfix

Настройка прямой DNS-зоны

```
[root@client ~]# dnf install s-nail
Last metadata expiration check: 1:39:11 ago on Sat 25 Oct 2025 12:07:43 PM UTC.
Package s-nail-14.9.22-6.el9.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[root@client ~]# cd /var
```

Рис. 11: Файл прямой DNS-зоны

Настройка обратной DNS-зоны

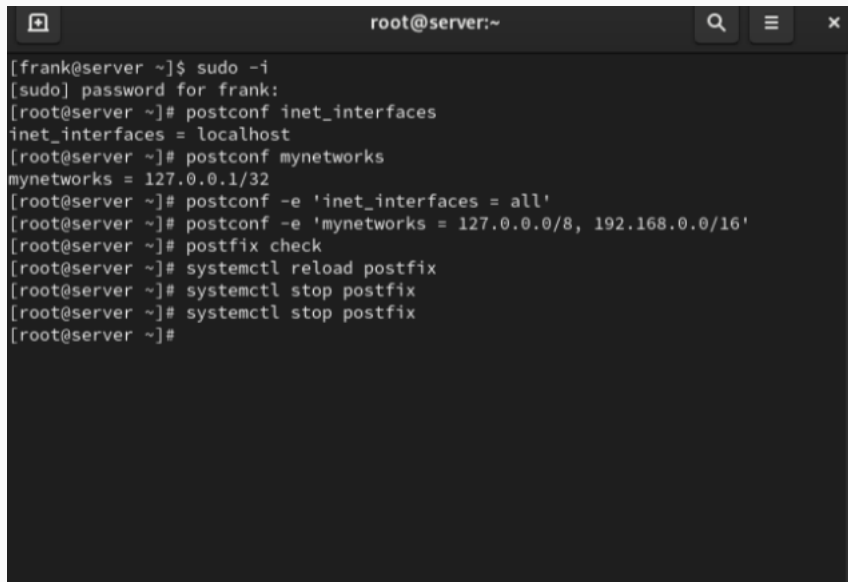
```
[root@client ~]# postconf inet_protocols
inet_protocols = all
[root@client ~]# postconf -e 'inet_protocols = ipv4'
[root@client ~]# systemctl enable postfix
Created symlink /etc/systemd/system/multi-user.target.wants/postfix.service → /usr/lib/systemd/system/postfix.service.
[root@client ~]# systemctl start postfix
[root@client ~]#
```

Рис. 12: Файл обратной DNS-зоны

Добавление домена в mydestination

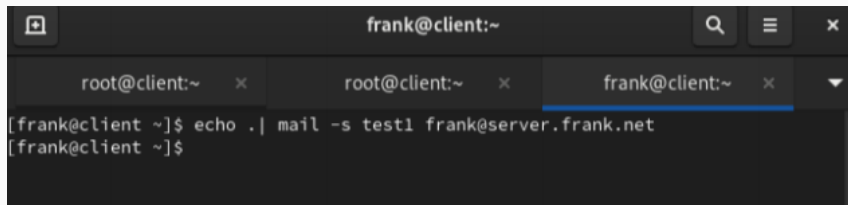
```
root@client:~  
[frank@client ~]$ sudo -i  
[sudo] password for frank:  
[root@client ~]# tail -f /var/log/maillog  
Oct 24 22:40:54 client postfix/cleanup[3460]: 62DE31185C18: message-id=<20251024  
224054.62DE31185C18@client.user.net>  
Oct 24 22:40:54 client postfix/qmgr[3459]: 62DE31185C18: from=<root@client.user.  
net>, size=321, nrcpt=1 (queue active)  
Oct 24 22:40:54 client postfix/smtp[3462]: warning: run-time library vs. compile  
-time header version mismatch: OpenSSL 3.0.0 may not be compatible with OpenSSL  
3.2.0  
Oct 24 22:40:54 client postfix/smtp[3462]: 62DE31185C18: to=<frank@frank.user.ne  
t>, relay=none, delay=204, delays=204/0.07/0.27/0, dsn=5.4.4, status=bounced (Ho  
st or domain name not found. Name service error for name=frank.user.net type=A:  
Host not found)  
Oct 24 22:40:54 client postfix/cleanup[3460]: CD15C1185C19: message-id=<20251024  
224054.CD15C1185C19@client.user.net>  
Oct 24 22:40:54 client postfix/qmgr[3459]: CD15C1185C19: from=<>, size=2359, nrc  
pt=1 (queue active)  
Oct 24 22:40:54 client postfix/bounce[3464]: 62DE31185C18: sender non-delivery n  
otification: CD15C1185C19  
Oct 24 22:40:54 client postfix/qmgr[3459]: 62DE31185C18: removed  
Oct 24 22:40:54 client postfix/local[3465]: CD15C1185C19: to=<root@client.user.n  
et>, relay=local, delay=0.02, delays=0.01/0.01/0/0.01, dsn=2.0.0, status=sent (d  
elivered to mailbox)
```

Успешная доставка доменной почты

A terminal window titled 'root@server:~' with standard window controls (minimize, maximize, close) and search, menu, and close buttons. The terminal shows a user 'frank' running 'sudo -i' to become root. Then, 'postconf inet_interfaces' is run, showing 'localhost'. Next, 'postconf mynetworks' is run, showing '127.0.0.1/32'. Then, 'postconf -e 'inet_interfaces = all'' and 'postconf -e 'mynetworks = 127.0.0.0/8, 192.168.0.0/16'' are run to update the configuration. Finally, 'postfix check', 'systemctl reload postfix', and 'systemctl stop postfix' are run twice. The prompt returns to root@server ~# after each command.

```
[frank@server ~]$ sudo -i
[sudo] password for frank:
[root@server ~]# postconf inet_interfaces
inet_interfaces = localhost
[root@server ~]# postconf mynetworks
mynetworks = 127.0.0.1/32
[root@server ~]# postconf -e 'inet_interfaces = all'
[root@server ~]# postconf -e 'mynetworks = 127.0.0.0/8, 192.168.0.0/16'
[root@server ~]# postfix check
[root@server ~]# systemctl reload postfix
[root@server ~]# systemctl stop postfix
[root@server ~]# systemctl stop postfix
[root@server ~]#
```

Подтверждение письма в почтовом ящике



A terminal window with a dark background. The title bar shows 'frank@client:~' and search, menu, and close icons. The tab bar has three tabs: 'root@client:~', 'root@client:~', and 'frank@client:~' (which is selected and highlighted with a blue underline). The terminal content shows the command `[frank@client ~]$ echo . | mail -s test1 frank@server.frank.net` and the prompt `[frank@client ~]$` on the next line.

```
frank@client:~  
[frank@client ~]$ echo . | mail -s test1 frank@server.frank.net  
[frank@client ~]$
```

Рис. 15: Полученное доменное письмо

Выводы по проделанной работе

- Установлен и настроен Postfix на виртуальных машинах server и client
- Выполнена настройка параметров Postfix через postconf и проверка конфигурации
- Подтверждена доставка локальной почты и пересылка между узлами сети
- Реализована доставка на доменные адреса через настройку DNS-зон и mydestination
- Подготовлены provisioning-скрипты mail.sh для автоматизации развёртывания